LESSON PLAN

Cyber Security Awareness Workshop

Facilitator delivery and evidence plan

Course Code: C506
Version 1.0 | 1 October 2026

Course information

Field	Value
Course	Cyber Security Awareness Workshop
Code	C506
Duration	1 day, 9:30am – 5:30pm (7.5 instructional hours, 0.5 hour lunch)
Slides	139

Document Version Control Record

Version	Effective date	Summary	Author
1.0	1 Oct 2026	Initial release of the workshop courseware	Tertiary Infotech Academy

Learning outcomes

Outcome	Statement	Topic
LO1	Identify cyber security risks and threats affecting people and organisations	Topic 1
LO2	Apply safeguards to protect personal information and everyday digital activity	Topic 2
LO3	Apply business safeguards and interpret incidents that require escalation	Topic 3

Facilitation principles

- Use the PPT for concepts, visuals, discussion and lab framing; use the LG/lab pack for detailed procedures.
- Show the mock communication first and ask ‘What would you do next?’ before revealing the clue cards.
- Ask learners to separate evidence from inference and link every control to a specific risk.
- Use only supplied mock data and authorised training environments.
- Do not demonstrate destructive techniques, credential theft or testing outside scope.
- Collect evidence against the acceptance criteria rather than screenshots alone.

Detailed schedule

Time	Duration	Topic / activity	Materials	Evidence / method	Outcome
09:30–09:45	15 min	Welcome, outcomes and safety	Slides 1–4	Questioning + scope acknowledgement	All
09:45–10:20	35 min	Core model, CIA, risk and Singapore BEC data	Slides 5–10	Asset–threat–vulnerability chain	LO1
10:20–11:00	40 min	Threat actors, attack chain and malware	Topic 1 slides	Malware classification discussion	LO1
11:00–11:40	40 min	Social engineering through eight mock emails, messages, calls, QR and workplace examples	Beginner example slides + LG	Spot-the-clue discussion and safe-response choices	LO1/LO3
11:40–12:20	40 min	Real-world cases: SingHealth through BEC	Case slides	Attack-chain transfer questions	LO1/LO3
12:20–12:55	35 min	Labs 01–03: triage, phishing/BEC, malware	Lab folders 01–03	Evidence checkpoint	LO1/LO3

Time	Duration	Topic / activity	Materials	Evidence / method	Outcome
12:55–13:25	30 min	Lunch break	—	—	—
13:25–14:05	40 min	Personal safeguards: identity, device, backup, online safety	Topic 2 slides	Personal baseline self-audit	LO2
14:05–14:40	35 min	Lab 06: cryptography safeguards	Lab 06	Hash/encryption evidence	LO2
14:40–15:20	40 min	Business vulnerabilities and risk prioritisation	Topic 3 slides	Risk-register discussion	LO1/LO3
15:20–16:25	65 min	Labs 04, 05, 07 and 08	Lab folders	Facilitator evidence review	LO1–LO3
16:25–17:00	35 min	Detection, governance, recovery and escalation	Topic 3 slides	Incident-brief rehearsal	LO3
17:00–17:30	30 min	Labs 09–10 and action plan	Lab folders 09–10	Three-minute briefing	LO1–LO3

Lab facilitation map

Lab	Time	Tool	Alignment	Evidence focus
Lab-01	25 minutes	Cybersecurity Simulator	LO1	Completed triage table; Severity rationale; Screenshot of highest-risk decision
Lab-02	35 minutes	Cybersecurity Simulator + Regex Generator	LO1, LO3	Annotated inbox; Tested regex; Detection test results
Lab-03	30 minutes	Cybersecurity Simulator	LO1, LO3	Malware classification; Attack-chain map; Evidence vs inference notes
Lab-04	35 minutes	Ethical Hacking Simulator	LO1	Ranked risk register; Top-three rationale; Treatment decisions
Lab-05	40 minutes	FauxBank	LO1, LO3	Signed scope check; Completed test plan; Redacted screenshots
Lab-06	30 minutes	Cryptography Toolkit	LO2, LO3	Data classification; Hash comparison; Encrypted/decrypted sample
Lab-07	35 minutes	IP Calculator	LO1, LO3	Subnet plan; Asset-to-zone mapping; Allowed-flow matrix
Lab-08	35 minutes	PCAP Analyzer	LO1, LO3	Traffic findings; Five-event timeline; Earliest reliable signal
Lab-09	30 minutes	Regex Generator	LO3	Tested regex; Confusion matrix; Threshold decision
Lab-10	45 minutes	Integrated toolkit	LO1–LO3	Correlated timeline; CIA and business-impact assessment; Decision log

Real-world case prompts

SingHealth data breach

Ask: What was the entry path? Which weakness made it credible? What could defenders see? Which earlier control could have broken the chain? Transfer the answer to a learner's own context. Source: Ministry of Health Singapore — SingHealth's IT System Target of Cyberattack — https://www.moh.gov.sg/newsroom/singhealth%27s-it-system-target-of-cyberattack/

WannaCry and the NHS

Ask: What was the entry path? Which weakness made it credible? What could defenders see? Which earlier control could have broken the chain? Transfer the answer to a learner's own context. Source: UK National Audit Office —

Investigation: WannaCry Cyber Attack and the NHS — https://www.nao.org.uk/reports/investigation-wannacry-cyber-attack-and-the-nhs/

Equifax breach

Ask: What was the entry path? Which weakness made it credible? What could defenders see? Which earlier control could have broken the chain? Transfer the answer to a learner's own context. Source: U.S. House Committee on Oversight — The Equifax Data Breach — https://oversight.house.gov/report/the-equifax-data-breach/

SolarWinds supply-chain compromise

Ask: What was the entry path? Which weakness made it credible? What could defenders see? Which earlier control could have broken the chain? Transfer the answer to a learner's own context. Source: CISA — Active Exploitation of SolarWinds Software — https://www.cisa.gov/news-events/alerts/2020/12/13/active-exploitation-solarwinds-software

Colonial Pipeline

Ask: What was the entry path? Which weakness made it credible? What could defenders see? Which earlier control could have broken the chain? Transfer the answer to a learner's own context. Source: CISA — IAM Case Study: Colonial Pipeline — https://www.cisa.gov/sites/default/files/2023-12/IAM_EDUCATIONAL_AID_TALKING_POINTS_508C.pdf

Log4Shell exploitation

Ask: What was the entry path? Which weakness made it credible? What could defenders see? Which earlier control could have broken the chain? Transfer the answer to a learner's own context. Source: CISA — Mitigating Log4Shell and Log4j Vulnerabilities — https://www.cisa.gov/news-events/cybersecurity-advisories/aa21-356a

MOVEit mass exploitation

Ask: What was the entry path? Which weakness made it credible? What could defenders see? Which earlier control could have broken the chain? Transfer the answer to a learner's own context. Source: CISA — CL0P Exploits MOVEit Vulnerability — https://www.cisa.gov/sites/default/files/2023-06/aa23-158a-stopransomware-cl0p-ransomware-gang-exploits-moveit-vulnerability_7.pdf

MGM Resorts cyber incident

Ask: What was the entry path? Which weakness made it credible? What could defenders see? Which earlier control could have broken the chain? Transfer the answer to a learner's own context. Source: U.S. Securities and Exchange Commission — MGM Resorts International Form 8-K — https://www.sec.gov/Archives/edgar/data/789570/000119312523251667/d461062d8k.htm

Singapore commodity-trader BEC

Ask: What was the entry path? Which weakness made it credible? What could defenders see? Which earlier control could have broken the chain? Transfer the answer to a learner's own context. Source: Singapore Police Force — Recovery of Over USD 6.6 Million in BEC Case — https://www.police.gov.sg/media-hub/news/2026/05/20260505_international_cooperation_leads_to_the_recovery_of_over

Target payment-card breach

Ask: What was the entry path? Which weakness made it credible? What could defenders see? Which earlier control could have broken the chain? Transfer the answer to a learner's own context. Source: U.S. Senate Committee on Commerce — Target's Missed Opportunities to Stop a Data Breach —

https://www.commerce.senate.gov/press/dem/release/rockefeller-staff-report-details-targets-missed-opportunities-to-stop-massive-data-breach-2014-3/

NotPetya disruption at Maersk

Ask: What was the entry path? Which weakness made it credible? What could defenders see? Which earlier control could have broken the chain? Transfer the answer to a learner's own context. Source: UK Department for Transport — Cyber Security for Ports and Port Systems — https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/546160/cyber-security-for-ports-and-port-systems-code-of-practice.pdf

Marriott Starwood data breach

Ask: What was the entry path? Which weakness made it credible? What could defenders see? Which earlier control could have broken the chain? Transfer the answer to a learner's own context. Source: UK Information Commissioner's Office — Marriott International Monetary Penalty Notice — https://ico.org.uk/media/action-weve-taken/mpns/2618524/marriott-international-inc-mpn-20201030.pdf

Capital One cloud data breach

Ask: What was the entry path? Which weakness made it credible? What could defenders see? Which earlier control could have broken the chain? Transfer the answer to a learner's own context. Source: U.S. Department of Justice — Seattle Tech Worker Arrested for Data Theft — https://www.justice.gov/usao-wdwa/pr/seattle-tech-worker-arrested-data-theft-involving-large-financial-services-company

Norsk Hydro LockerGoga ransomware

Ask: What was the entry path? Which weakness made it credible? What could defenders see? Which earlier control could have broken the chain? Transfer the answer to a learner's own context. Source: UK National Cyber Security Centre — What Is OT Malware? — https://www.ncsc.gov.uk/blog-post/what-is-ot-malware

Change Healthcare cyberattack

Ask: What was the entry path? Which weakness made it credible? What could defenders see? Which earlier control could have broken the chain? Transfer the answer to a learner's own context. Source: U.S. Securities and Exchange Commission — UnitedHealth Group Form 8-K, 21 February 2024 — https://www.sec.gov/Archives/edgar/data/731766/000073176624000045/unh-20240221.htm

Trainer close-out checklist

- ☐ All outcomes covered
- ☐ All lab safety boundaries stated
- ☐ Learner evidence checked against acceptance criteria
- ☐ Incidents or near misses documented
- ☐ Follow-up actions assigned